

PROJECT PLANNING & WBS

OptiCrop: Smart Agricultural Production Optimization Engine

Date	03 July 2026
Team ID	SWTID-2026-7253
Team Size	5
Team Leader	Bommidi Deepika
Team Members	G Uday Kumar, Kavali Anu, Pavan Kumar Sowdepalli, Dasarayyagari Sravan Kumar
Maximum Marks	4 Marks

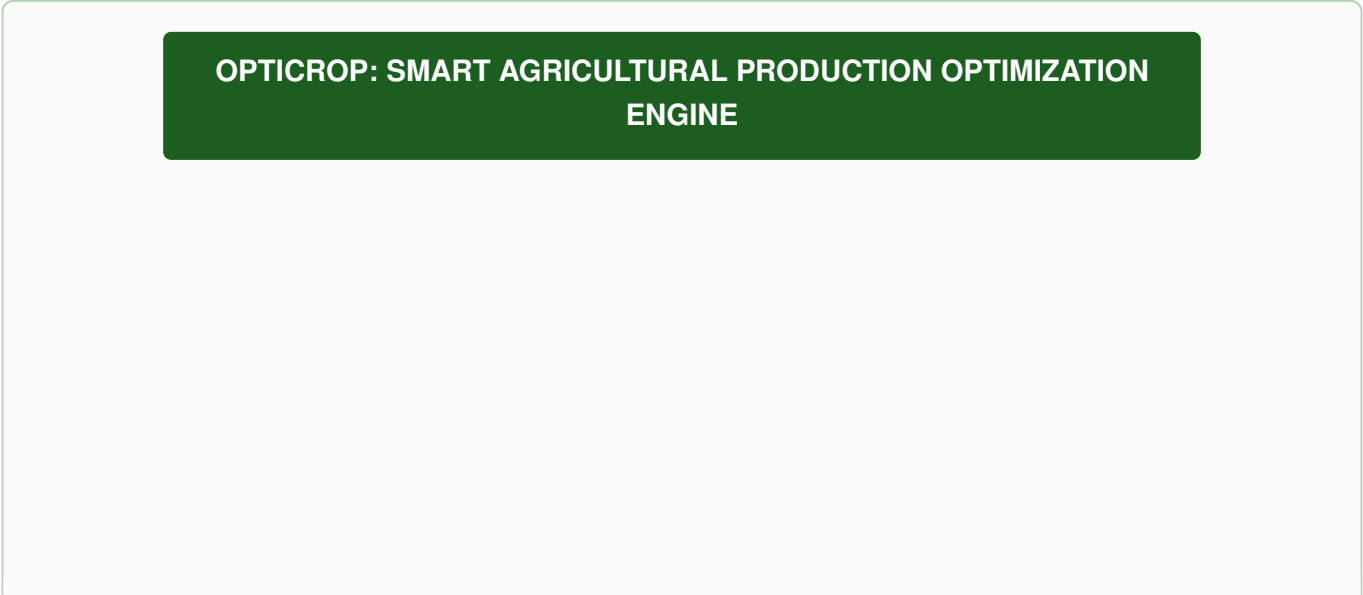
Introduction

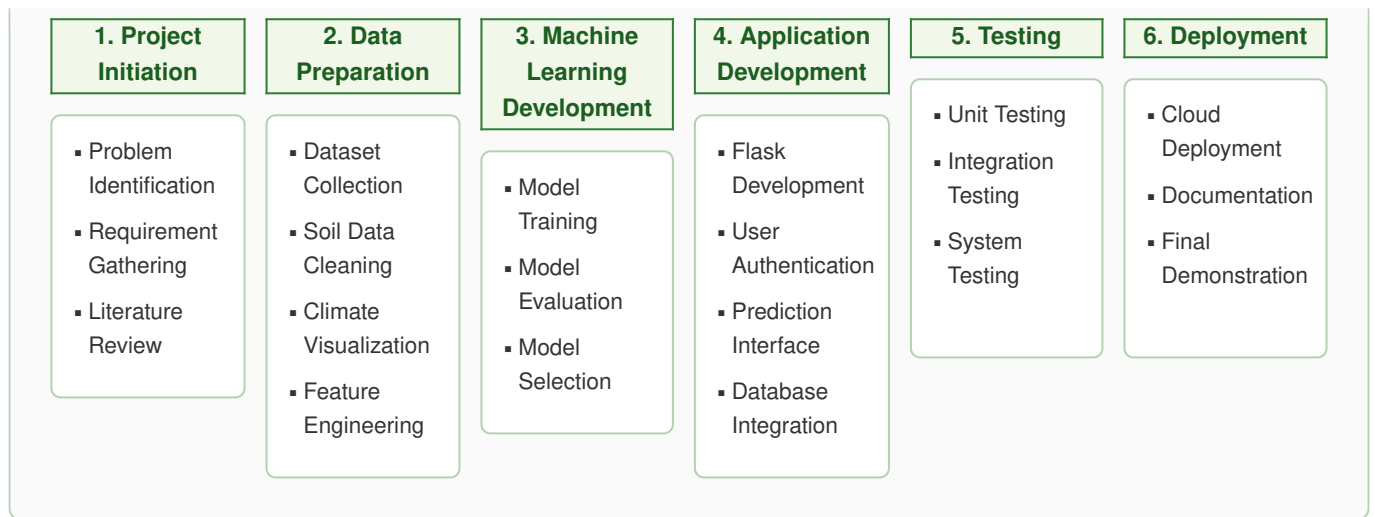
The Project Planning document defines the overall execution strategy for developing the **OptiCrop: Smart Agricultural Production Optimization Engine**. It identifies project phases, major activities, deliverables, team responsibilities, and the planned development sequence required for successful project completion.

Project Objectives

- Develop a secure Machine Learning-based crop production optimization engine.
- Automate soil compatibility and regional crop eligibility evaluation workflows.
- Achieve high predictive matching metrics using advanced multi-class supervised learning models.
- Build a fast, responsive Flask web application adapted for field and remote environment deployment.
- Store granular environmental history matrices and farmers' prediction analytics profiles securely.
- Deploy a highly scalable web microservice solution suitable for regional agricultural centers and policy tracking.

Work Breakdown Structure (WBS)





Conclusion

The Work Breakdown Structure (WBS) provides a systematic approach for executing the project by dividing the complete development process into manageable, logical phases. This systematic organization optimizes timeline planning, enhances cross-functional team coordination, and facilitates continuous progress monitoring to ensure the timely deployment of the OptiCrop engine.